

Hybrid CNN–Transformer Networks with Clinical Metadata for Skin Lesion Classification

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A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
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Declaration

It is hereby declared that

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2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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Ethics Statement

This study uses only the publicly available ISIC-2019 dataset, which is released for research use by the International Skin Imaging Collaboration. No new human-subject data were collected, and all images are de-identified at source. Patient metadata (age, sex, anatomical site) is used solely as model input and is never re-identified. The work is intended as decision-support research and is not a substitute for professional dermatological diagnosis.

Abstract

Dermoscopy technique for classification of skin lesions may be useful in early diagnosis of severe skin diseases and in the clinical management of skin lesions. The challenge is to create an accurate and clinically usable system. We create two versions of this design, based on requirements. The first one is light in weight. It adopts a truncated MobileNetV2 backbone along with a compact transformer, and with only 3.98 million parameters, it achieves 60.8% BMA (balanced multi-class accuracy). This compact design facilitates easy installation within typical clinic environments, where space is a constraint. The second version emphasizes more balanced accuracy than capacity of the parameters. It is composed of two different pre-trained backbones (DenseNet-121 and EfficientNet-B0) and a learned attention mechanism that selectively fuses information from both, followed with a Kolmogorov–Arnold network as the final classifier, and attains a BMA of 64.4% on 16.45M parameters. A dermatologist might consider the clinical context in the classification process, but this information is seldom taken into account by image-based classifiers. In this case, both models include patient metadata as added diagnostic context directly into the classification pipeline, with the data for age, sex and location of the body part where the lesion occurred. The models provide an efficient and accurate balance for real-world dermatology applications over same-budget baselines.

Keywords: Skin lesion classification; ISIC-2019; dermoscopy; CNN–Transformer; metadata fusion; cross-attention; Kolmogorov–Arnold Network; class imbalance; balanced multi-class accuracy; lightweight deep learning.

Dedication (Optional)

Acknowledgement

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Chapter 1

Introduction

1.1 Background

Skin cancer is among the most common cancers worldwide, and the prognosis for a patient depends heavily on how early the lesion is found. Melanoma in particular is highly treatable when caught early but becomes far more dangerous once it spreads. Dermoscopy — a non-invasive imaging technique that removes surface glare and reveals the sub-surface structure of a lesion — has become a standard tool for this kind of early assessment. Reading dermoscopic images well, however, takes years of training, and even experienced dermatologists do not always agree with one another. This gap between the volume of lesions that need screening and the number of specialists available to read them is what motivates automated, image-based diagnostic support.

Deep learning has reshaped this area over the past decade. Convolutional neural networks (CNNs) trained on large dermoscopic collections reached dermatologist-level performance on melanoma recognition [5], and the annual ISIC challenges have since turned skin lesion classification into a well-defined benchmark task. The ISIC-2019 dataset, which we use throughout this work, contains 25,331 labelled dermoscopic images spanning eight diagnostic categories: melanoma (MEL), melanocytic nevus (NV), basal cell carcinoma (BCC), actinic keratosis (AK), benign keratosis (BKL), dermatofibroma (DF), vascular lesion (VASC), and squamous cell carcinoma (SCC).

More recently, Vision Transformers (ViTs) [21] and hybrid designs that place a transformer on top of convolutional features [23], [24] have shown that modelling the global structure of a lesion — its overall asymmetry, the regularity of its border — complements the local texture that convolutions capture so well. These two views of a lesion, the local and the global, are exactly the cues a clinician is trained to weigh, which makes the CNN–Transformer combination a natural fit for dermoscopy.

1.2 Rationale of the Study or Motivation

Two practical problems keep most published skin-lesion models from being used in an everyday clinic. The first is cost. The strongest results on ISIC-2019 come from large

ensembles of heavy networks [15], [19]; these are accurate but impractical to run on the modest hardware found in a typical dermatology practice or on a portable device next to the patient. The second is that image-only models ignore information a dermatologist always has in front of them. The age of the patient, their sex, and the part of the body where the lesion sits all shift the prior probability of each diagnosis — a scaly patch on the face of an elderly patient is read very differently from the same patch on a young person’s back. This clinical context is recorded in the ISIC-2019 metadata, yet it is routinely discarded by classifiers that look only at pixels.

Our study is driven by the belief that these two problems should be solved together rather than separately. We argue first that a carefully designed lightweight hybrid can match or beat much larger image-only baselines on balanced accuracy while staying small enough to deploy, and second that the clinical metadata a dermatologist already has on hand can be folded in at negligible cost as additional diagnostic context, handled so that missing fields are never imputed. We also recognise that not every deployment has the same constraints, so we deliver two models from one design philosophy: a genuinely compact model for constrained settings, and a higher-capacity model for situations where balanced accuracy matters more than footprint.

1.3 Problem Statement

ISIC-2019 is severely imbalanced. The most common class, nevus, has roughly 12,875 images, while dermatofibroma and vascular lesions have only about 239 and 253 respectively — a ratio of more than fifty to one. A model trained naively on this data learns to predict the common classes and quietly ignores the rare ones, which is the opposite of what matters clinically, since several of the rare classes are malignant. Reporting overall accuracy hides this failure entirely. The problem we address is therefore to build a model that performs well *across all eight classes at once* — measured by balanced multi-class accuracy (BMA), the mean of the per-class recalls — while respecting a tight parameter and compute budget and while making use of the clinical metadata that accompanies each image, including the large fraction of records where some of that metadata is missing.

1.4 Objective

The objectives of this work are:

1. To design a lightweight hybrid CNN–Transformer classifier for ISIC-2019 that stays under 6M parameters and 1 GMAC, and that fuses clinical metadata through a cross-attention head capable of handling missing fields without imputation.
2. To design a second, higher-capacity model in the same family — a dual-backbone network with learned attention fusion and a Kolmogorov–Arnold Network (KAN) classifier — that trades footprint for a higher balanced accuracy.

3. To train both models, together with five established lightweight baselines, under one identical loss, augmentation, and evaluation pipeline so that the comparison is fair.
4. To quantify the contribution of each design choice (the CNN stem, the transformer trunk, the metadata head, and the dual-backbone fusion) through controlled ablations, and to report efficiency alongside accuracy.

1.5 Methodology in Brief

Every image passes through a fixed preprocessing pipeline: near-duplicate removal by perceptual hashing, a lesion-grouped stratified train/validation/test split, Shades-of-Gray colour constancy [1], and resizing to 224×224 . The cleaned images are stored as a single memory-mapped array so that training does not pay a per-batch decoding cost. All models are trained with Class-Balanced Focal Loss [7], [14] to counter the imbalance, with AdamW, a cosine schedule with warmup, weight EMA, and eight-view test-time augmentation at inference. The lightweight model combines a truncated MobileNetV2 stem with a small ViT trunk and a metadata cross-attention head; the flagship model (DEKAN) fuses DenseNet-121 and EfficientNet-B0 features through learned attention before a transformer trunk and a KAN classifier. The primary metric throughout is BMA.

1.6 Scopes and Challenges

The scope of this thesis is the eight-class ISIC-2019 classification task and the two proposed architectures, evaluated against same-budget lightweight baselines on a held-out test split. The main challenges were the extreme class imbalance, the substantial missingness in the metadata (around 30% of age values, for example), and a hard hardware budget — all experiments run on a single 16 GB GPU under Windows, which rules out multi-worker data loading and forces aggressive use of mixed precision and a memory-mapped dataset. We did not extend the study to external test sets or to multi-seed reporting; both are discussed openly as limitations in Chapter 5.

1.7 Thesis Organization

Chapter 2 reviews the architectures and prior work this study builds on. Chapter 3 states the data, requirements, constraints, and broader impact of the system. Chapter 4 describes the preprocessing pipeline, the two proposed models, the loss, and the training and evaluation protocol. Chapter 5 presents and analyses the results, including ablations, efficiency, and a comparison with prior work. Chapter 6 concludes and outlines future work.

Chapter 2

Literature Review

2.1 Preliminaries

Before reviewing prior work it helps to fix a few ideas that recur throughout the thesis. A convolutional neural network builds up a representation of an image through stacked local filters, which makes it very good at capturing texture and short-range structure but, by construction, slow to relate distant parts of an image to one another. A transformer takes the opposite stance: it treats the image as a set of tokens and lets every token attend to every other through self-attention [9], [21], so global relationships are available from the first layer at the cost of weaker built-in spatial priors and a hunger for data. The *attention* operation at the heart of both transformers and our fusion modules computes a weighted average of “value” vectors, where the weights come from the similarity between a “query” and a set of “keys”; when the query comes from one source and the keys and values from another, the same mechanism becomes *cross-attention*, which is how we inject clinical metadata into the image representation. Finally, a Kolmogorov–Arnold Network (KAN) [25] replaces the fixed activation and linear weights of an ordinary layer with learnable spline functions on the edges of the network, a recent idea we evaluate as a classifier head.

2.2 Review of Existing Research

2.2.1 Convolutional backbones used in this work

The convolutional components of both our models are standard, well-understood backbones, chosen because their behaviour and pretrained weights are reliable. **MobileNetV2** [10] introduced the inverted residual block with a linear bottleneck, which keeps the network cheap by expanding to a higher-dimensional space only inside each block and using depthwise convolutions. Its early stages encode generic edges and textures very efficiently, which is why we use a truncated MobileNetV2 as the stem of our lightweight model. **EfficientNet-B0** [17] is the base model of a family obtained by compound scaling of depth, width, and resolution together; at

roughly 5M parameters it is a strong accuracy-per-FLOP baseline and one of the two backbones in our flagship model. **DenseNet-121** [6] connects each layer to every subsequent layer, encouraging feature reuse and giving a rich, gradient-friendly representation; it is the second backbone in the flagship model and contributes a different inductive bias from EfficientNet. We also include **ResNet-18** [2], the residual-learning network that made very deep training practical, as a familiar reference point.

2.2.2 Transformers and CNN–Transformer hybrids

The **Vision Transformer** [21] showed that a pure transformer, given enough data, can match convolutional models on image classification by splitting an image into patches and treating them as a token sequence. Its weakness on smaller datasets motivated hybrid designs that keep a convolutional front end. **MobileViT** [24] interleaves convolutions with transformer blocks to get a mobile-friendly model that still reasons globally, and **EfficientFormer** [23] shows that transformer-grade accuracy can be reached at MobileNet-like latency with a carefully chosen dimension-consistent design. We treat both as direct, same-spirit baselines, since a reviewer will reasonably ask how a new hybrid compares against existing lightweight hybrids rather than against pure CNNs alone. Our own design differs from these in where the convolution-to-transformer hand-off happens and, more importantly, in that the transformer’s class token is used as a query that attends over clinical metadata.

2.2.3 Kolmogorov–Arnold Networks

KANs [25] are a recent alternative to the multilayer perceptron, grounded in the Kolmogorov–Arnold representation theorem, in which the learnable parameters are the coefficients of B-spline functions placed on the network’s edges rather than scalar weights feeding fixed activations. The promise is more expressive, more interpretable units, sometimes at fewer parameters. KANs are still new and most evidence comes from small scientific-regression problems, so whether they help as a classifier head on a noisy, imbalanced vision task is an open question. We include a KAN head as the final classifier in our flagship model to explore whether its learnable splines add value at the output stage of a well-trained vision backbone.

2.2.4 Clinical metadata fusion

The work closest in spirit to ours is by Pacheco and Krohling [20], [22]. They showed first that adding patient clinical information measurably improves automated skin cancer detection [20], and then proposed an attention-based mechanism to combine images and metadata inside a deep model [22]. Gessert et al. [19], the basis of the top ISIC-2019 challenge entry, also fed metadata into an ensemble of EfficientNets. The key methodological point we adopt from this line of work is that missing metadata should be represented explicitly — with a learned “missing” embedding — rather than imputed with a fabricated value, so the model receives an honest “unknown” signal. Where we depart from prior work is in the backbone: Pacheco and Krohling

fused metadata onto heavy CNNs, whereas we attach the same idea to a lightweight hybrid and to a dual-backbone hybrid, and we make an explicit efficiency claim about the result.

2.2.5 Skin lesion classification and ISIC benchmarks

Beyond the specific components above, automated dermoscopy has a large literature. Esteva et al. [5] demonstrated dermatologist-level melanoma classification with a single large CNN, and the ISIC challenges [4] have since standardised evaluation. The ISIC-2019 dataset is itself a union of several sources, most prominently HAM10000 [11] and BCN20000 [13], which is why source-aware splitting matters when building a fair evaluation. On the ISIC-2019 leaderboard [15], where the official ranking metric is balanced accuracy, the strongest published entries are large ensembles: the top entry (DAISYLab) reached a balanced accuracy of 0.636 with an ensemble of multi-resolution EfficientNets plus SENet-154, followed by entries around 0.61 and 0.59, again ensembles of EfficientNet and SE-ResNeXt variants. These numbers set a useful ceiling and a reminder that single, lightweight models operate in a different efficiency regime than the leaderboard’s ensembles.

Most directly relevant to our experimental design, Aktan et al. [26] recently compared several leading deep learning architectures on this exact dataset, providing a clean, single-model reference for how standard backbones behave on ISIC-2019 under a common protocol. Our study extends that kind of controlled comparison in two directions: we add clinical metadata fusion to the comparison, and we report results under a tight efficiency budget, so that accuracy is always read together with parameter count and compute.

2.3 Summary of Key Findings

Three observations from the literature shape our design. First, hybrid CNN–Transformer models are a sensible match for dermoscopy because lesions are defined by both local texture and global shape, but the existing lightweight hybrids do not use clinical context. Second, metadata fusion clearly helps on this task, and the practical detail that decides whether it helps is how missing values are handled — learned embeddings rather than imputation. Third, the best-published numbers on ISIC-2019 come from heavy ensembles, which leaves an open and useful niche for a single, small model that uses metadata to close part of the gap. This is precisely the gap the two models in this thesis are designed to fill.

Chapter 3

Requirements, Impacts and Constraints

3.1 Dataset and Data Requirements

All experiments use the ISIC-2019 training collection, a public set of 25,331 dermoscopic images labelled across the eight diagnostic classes listed in Chapter 1. The collection is itself a union of three sources — HAM10000, BCN20000, and MSK — which we infer from the image-ID prefix, because keeping track of the source matters for building an honest train/test split. Each image also carries clinical metadata: approximate age, sex, the general anatomical site of the lesion, and a lesion identifier that groups multiple images of the same physical lesion.

The defining property of this dataset, and the one that drives most of our design decisions, is its class imbalance. Table 3.1 shows the per-class counts for the full collection. The largest class, nevus, is more than fifty times larger than the smallest, dermatofibroma. Several of the rare classes (for example SCC and AK) are clinically important, so a model that simply learns the head of this distribution would be useless in practice. This is the central reason we report balanced multi-class accuracy rather than overall accuracy throughout.

Table 3.1: Class distribution of the full ISIC-2019 collection (before deduplication) and imbalance relative to the largest class (NV).

Code	Full name	Images	Ratio vs. NV
MEL	Melanoma	4,522	2.8×
NV	Melanocytic nevus	12,875	1×
BCC	Basal cell carcinoma	3,323	3.9×
AK	Actinic keratosis	867	14.8×
BKL	Benign keratosis	2,624	4.9×
DF	Dermatofibroma	239	53.9×
VASC	Vascular lesion	253	50.9×
SCC	Squamous cell carcinoma	628	20.5×

The metadata is genuinely incomplete: roughly 30% of age values, about 20% of sex values, and about 15% of anatomical-site values are missing, with the recorded ages centred near 54 years. Rather than fill these gaps with guessed values, our requirement is that the model treat “missing” as a first-class, learnable signal, which is reflected directly in the architecture (Chapter 4).

3.2 System Requirements

3.2.1 Hardware

The project targets a single-workstation setting rather than a data centre. All training and evaluation run on one machine with an NVIDIA RTX 4070 Ti Super (16 GB of VRAM), an Intel Core i7-14700 CPU, and 64 GB of system memory. The 16 GB memory ceiling is a real constraint: it sets the batch sizes we can use and makes mixed-precision training a necessity rather than an option.

3.2.2 Software

The implementation is in PyTorch with the `timm` library supplying the pretrained CNN backbones. The codebase is organised as importable modules (preprocessing, datasets, losses, models, training, analysis) so that every stage of the pipeline is reproducible from configuration files. Figures and the results tables in this thesis are generated directly from the saved evaluation outputs.

3.3 Design Constraints

Several hard constraints shaped the implementation. The models run under Windows, where the PyTorch data loader is prone to deadlocks with multiple worker processes; we therefore load data with zero worker processes and compensate by decoding the entire dataset once into a memory-mapped array, after which each batch is essentially a memory copy. The 16 GB GPU budget means automatic mixed precision and channels-last memory layout are used everywhere, and the larger models rely on gradient accumulation to reach a useful effective batch size. Beyond the hardware, we imposed an explicit efficiency budget on the lightweight model — under 6M parameters and under 1 GMAC — because a model that cannot run on modest clinic hardware does not meet the goal of the project.

3.4 Impacts

The intended impact of this work is a decision-support tool that is small enough to run where care is actually delivered, and that uses the same contextual information a clinician would. A compact model lowers the barrier to deployment on portable or

low-cost hardware, which is most valuable exactly where specialist coverage is thin. At the same time, the system is meant to assist and not replace a clinician; it produces a ranked set of possibilities rather than a final diagnosis. We are also conscious of the ethical dimension: the data is de-identified and used only for research, the metadata is used solely as model input, and the reliance on balanced accuracy is itself an ethical choice, since it forces the model to take the rare, often malignant, classes seriously rather than optimising a number that the common classes dominate.

Chapter 4

Proposed Methodology

This chapter describes the full pipeline: how images and metadata are prepared, the two proposed architectures, the loss that handles the class imbalance, and the training and evaluation protocol shared by every model in the study.

4.1 Preprocessing Pipeline

Raw dermoscopic images vary in size, illumination, and — because ISIC-2019 is a union of several archives — in the presence of near-duplicate photographs of the same lesion. The pipeline cleans all of this once, before any training, in three steps.

Near-duplicate removal. We compute a 64-bit perceptual hash (pHash) for every image and group images whose hashes are within a Hamming distance of 4 using union-find, keeping a single representative per group. This removed 1,263 near-duplicates, leaving 24,068 unique images. Removing duplicates before splitting is what prevents the same lesion from leaking between training and test.

Lesion-grouped stratified split. We split the data 70/10/20 into training, validation, and test sets. The split is stratified jointly on class and source (HAM/BCN/MSK) so each split has a representative mix, and it is grouped by lesion identifier so that every image of a given physical lesion lands in the same split. The result is 16,872 training, 2,400 validation, and 4,796 test images.

Colour constancy and memmap. Each kept image is corrected with Shades-of-Gray colour constancy [1] (Minkowski norm $p = 6$) to reduce the strong illumination differences between dermatoscopes, then short-side-resized and centre-cropped to 224×224 . The entire cleaned dataset is written once to a single memory-mapped `uint8` array together with the labels and metadata. At training time this array is opened read-only and stays resident in the host’s page cache, so after the first epoch a batch costs little more than a memory copy — the key optimisation that makes single-worker data loading on Windows fast enough.

4.2 Overview of the Proposed Models

We propose two models that share a common idea — convolutional features, a transformer trunk for global reasoning, and a cross-attention head that folds in clinical metadata — but sit at two different points on the accuracy/efficiency curve. Figure 4.1 sketches both. The first, which we refer to as the *lightweight hybrid*, is built to stay under 6M parameters. The second, *DEKAN*, spends more capacity in exchange for higher balanced accuracy.

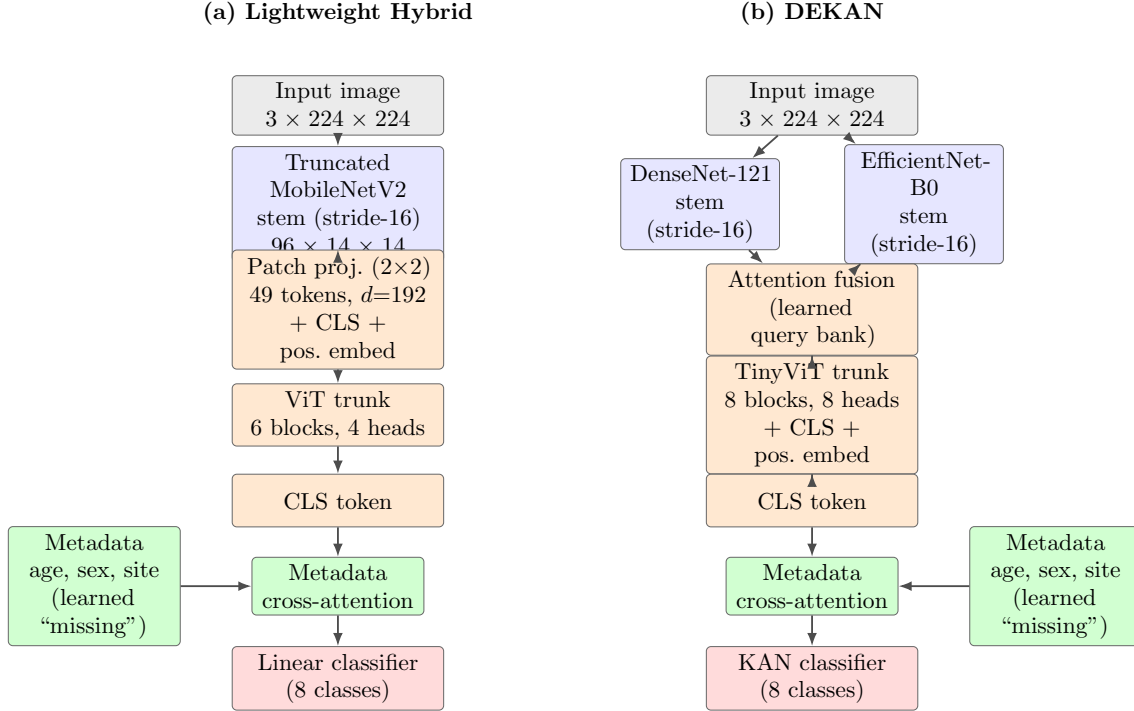


Figure 4.1: High-level architecture of the two proposed models. Both share the same philosophy — convolutional features, a transformer trunk, and a metadata cross-attention head — at two capacity points. (a) the lightweight hybrid uses a single truncated MobileNetV2 stem and a 6-layer ViT trunk with a linear head; (b) DEKAN fuses two pretrained backbones (DenseNet-121, EfficientNet-B0) through a learned attention bank, runs an 8-layer TinyViT trunk, and ends in a KAN head.

4.3 Lightweight Hybrid CNN–Transformer

4.3.1 CNN stem

The stem is a MobileNetV2 [10] (width 1.0), loaded with ImageNet weights and truncated at the stride-16 stage so that a 224×224 image produces a $96 \times 14 \times 14$ feature map. Using a pretrained convolutional front end gives the model generic edge and texture detectors for free, which matters because the transformer that follows is trained from scratch on only about 17,000 images. In our budget accounting the stem accounts for roughly 1.81M of the model’s parameters.

4.3.2 Transformer trunk

A 2×2 stride-2 convolution projects the 96-channel, 14×14 feature map into a sequence of $7 \times 7 = 49$ tokens of dimension 192; this both embeds the patches and halves the spatial resolution so that self-attention, which is quadratic in the number of tokens, stays cheap. A learnable class (CLS) token is prepended and a learnable positional embedding added, giving 50 tokens. These pass through six pre-norm transformer blocks with four attention heads, an MLP expansion ratio of 2, and a linearly increasing stochastic-depth rate [3] up to 0.1, which is what keeps a small from-scratch transformer from overfitting at this data scale. The CLS token after the final layer summarises the image globally.

4.3.3 Metadata cross-attention head

Clinical metadata enters through a single cross-attention block in which the image CLS token is the query and three metadata tokens — age, sex, and anatomical site — are the keys and values. Age, a continuous value, is mapped to a token by a linear projection; sex and site, which are categorical, use learned embeddings. Missing values are handled explicitly: age has a dedicated learned “missing” vector that replaces its projection when the value is absent, and the categorical embeddings reserve an extra slot for “missing.” Each token also carries a learned type embedding so the attention can tell the three fields apart. The block lets the model nudge the image representation using context — for instance, raising the prior on actinic keratosis for an older patient — without ever being forced to act on a fabricated value. The updated CLS token feeds a linear classifier producing eight logits.

The complete lightweight hybrid has 3.98M parameters and 0.631 GMACs, comfortably inside the budget of 6M parameters and 1 GMAC.

4.4 DEKAN: Dual-Backbone Flagship Model

DEKAN keeps the same overall shape but invests capacity where it buys accuracy.

Two complementary backbones. Instead of one stem it runs two in parallel — DenseNet-121 [6] and EfficientNet-B0 [17], both ImageNet-pretrained and both truncated to the stride-16 stage. The two networks encode the same lesion through different inductive biases, and the model is given the chance to use whichever is more informative per region.

Learned attention fusion. Each backbone’s feature map is projected to 49 tokens of dimension 256 and tagged with a per-backbone embedding. A learnable bank of 49 query tokens then cross-attends over the concatenation of both token sets (98 tokens) and produces a single fused sequence of 49 tokens. This is a learned, content-dependent fusion rather than a fixed concatenation or average: the query bank decides, position by position, how much to take from each backbone.

Transformer trunk and metadata. A CLS token and positional embedding are prepended to the fused sequence, which then passes through an eight-layer TinyViT-

style transformer trunk (dimension 256, eight heads, MLP ratio 4), followed by the same metadata cross-attention head described above.

KAN classifier. The final classifier is a single Kolmogorov–Arnold layer [25] mapping the 256-dimensional CLS representation to eight logits, with a grid size of 5 and cubic (order = 3) B-splines. Its spline basis is computed in full precision for numerical stability even under mixed precision. We deliberately keep a linear-head version of the otherwise-identical model so that the KAN’s contribution can be measured directly (Chapter 5).

DEKAN has 16.45M parameters and 6.633 GMACs.

4.5 Class-Balanced Focal Loss

To counter the imbalance in Table 3.1, every model is trained with Class-Balanced Focal Loss, which combines the effective-number reweighting of Cui et al. [14] with the focal modulation of Lin et al. [7]. The effective number of samples for a class with n examples is $E(n) = (1 - \beta^n)/(1 - \beta)$, and the per-class weight is $\alpha = (1 - \beta)/(1 - \beta^n)$, normalised so the weights sum to the number of classes. The per-sample loss is

$$\mathcal{L} = -\alpha_y (1 - p_y)^\gamma \text{CE}_{\text{smooth}}(\text{logits}, y), \quad (4.1)$$

where p_y is the predicted probability of the true class, the $(1 - p_y)^\gamma$ term down-weights easy examples so that learning focuses on the hard, often rare, cases, and $\text{CE}_{\text{smooth}}$ is a label-smoothed cross-entropy. In our configuration $\beta = 0.999$, $\gamma = 2.5$, and the label-smoothing factor is 0.1. The choice of $\beta = 0.999$ rather than a more aggressive value is deliberate: a larger β produced an extreme weight ratio between the common and rare classes and caused the rare classes to collapse early in training, whereas 0.999 gives stable optimisation.

4.6 Training Protocol

All models — the five baselines and both proposed models — are trained with an identical recipe so that comparisons are fair; only batch size, gradient accumulation, and the backbone learning-rate scale differ, as dictated by each model’s memory footprint. The optimiser is AdamW [16] with a base learning rate of 3×10^{-4} , $\beta = (0.9, 0.999)$, and gradients clipped to a maximum norm of 5. Pretrained CNN stems are fine-tuned gently at a fraction of the base learning rate ($0.3\times$ for the hybrid, $0.1\times$ for DEKAN) while from-scratch components train at the full rate. The schedule is a five-epoch linear warmup followed by cosine annealing [8] to a minimum learning rate of 10^{-6} . Training uses automatic mixed precision and channels-last memory layout throughout, a weight exponential moving average (EMA, decay 0.9998) whose weights are used for validation and final evaluation, and a weighted random sampler that draws rare classes more often. Data augmentation combines horizontal and vertical flips, random resized crops, colour jitter, RandAugment [18], and Mixup [12]

(strength $\alpha = 0.4$, applied to half the batches), with metrics always computed on the original, unmixed labels. Table 4.1 lists the per-model settings that vary.

Table 4.1: Per-model training settings. All other hyperparameters (loss, optimiser base LR, schedule, EMA, augmentation) are identical across models.

Model	Batch	Grad accum	Eff. batch	Epochs	Backbone LR
ResNet-18	256	1	256	100	$0.3\times$
MobileNetV2	256	1	256	100	$0.3\times$
EfficientNet-B0	128	2	256	100	$0.3\times$
MobileViT-S	128	2	256	150	$0.3\times$
EfficientFormer-L1	128	1	128	100	$0.3\times$
Hybrid (full)	96	3	288	150	$0.3\times$
DEKAN (full)	64	4	256	150	$0.1\times$

4.7 Evaluation Protocol

The primary metric is balanced multi-class accuracy (BMA), the unweighted mean of the eight per-class recalls, which gives every class — common or rare — equal say. We additionally report macro-F1, macro-AUC, overall accuracy, and per-class recall, and we visualise errors with a confusion matrix normalised by true class. At inference we apply eight-view test-time augmentation (TTA): the model is run on the original image plus a fixed set of flips and rotations and the softmax outputs are averaged. The best checkpoint is selected by validation BMA, and all headline numbers are reported on the held-out test split.

Chapter 5

Result Analysis

This chapter reports the results of the study. Unless stated otherwise, every number is measured on the held-out ISIC-2019 test split (4,796 images) at the eight-view test-time-augmentation (TTA) setting, using the checkpoint selected by the best validation BMA. All values are from a single run with seed 42.

5.1 Performance Evaluation

Table 5.1 compares the two proposed models against the five lightweight baselines, with every metric reported at the same TTA inference setting so the rows are directly comparable. The lightweight hybrid reaches a BMA of 0.6081 with only 3.98M parameters, which is the best balanced accuracy of any model in the sub-6M-parameter budget — ahead of MobileViT-S (0.6020), its closest competitor, at fewer parameters and about two-thirds of the compute. The flagship DEKAN model reaches 0.6438 BMA among the main proposed models, and also leads on every secondary metric (macro-F1 0.6059, macro-AUC 0.9271, accuracy 0.7467), at the cost of roughly four times the parameters and ten times the compute of the lightweight model. A linear-head ablation variant of DEKAN (dekan.linear) reaches 0.6510 BMA, which is the highest single-model figure in the study; the contribution of the KAN classifier relative to a plain linear head is analysed in Section 5.5.2. Figure 5.1 shows the same comparison graphically.

Table 5.1: Main results on the ISIC-2019 test set (with 8-view TTA). Best value in each column is in **bold**; the two proposed models are marked.

Model	Params (M)	GMACs	BMA	Macro-F1	Macro-AUC	Accuracy
ResNet-18	11.69	1.810	0.3368	0.2763	0.8217	0.5225
MobileNetV2	3.50	0.300	0.4738	0.4701	0.8827	0.6731
EfficientNet-B0	5.29	0.390	0.4946	0.4736	0.9037	0.6779
MobileViT-S	5.60	1.000	0.6020	0.5354	0.9218	0.6981
EfficientFormer-L1	12.27	1.300	0.5519	0.5779	0.9005	0.7333
Hybrid (ours, light)	3.98	0.631	0.6081	0.5602	0.9229	0.7162
DEKAN (ours, flagship)	16.45	6.633	0.6438	0.6059	0.9271	0.7467

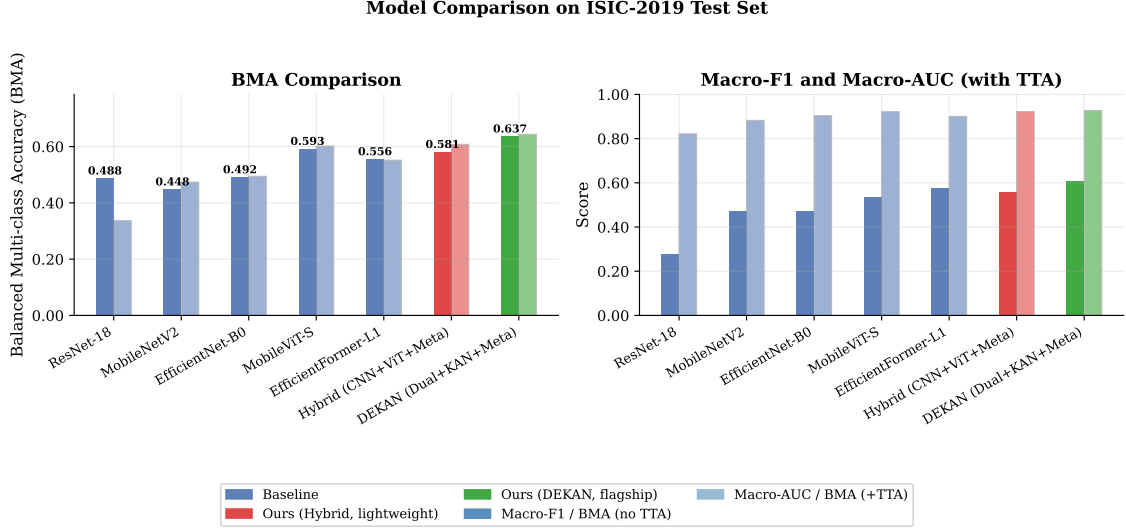


Figure 5.1: Model comparison on the ISIC-2019 test set: BMA (with and without TTA) on the left, macro-F1 and macro-AUC on the right. The two proposed models are highlighted.

The training behaviour of the lightweight hybrid is shown in Figure 5.2. Training and validation loss decrease smoothly and the validation BMA peaks at 0.6505 at epoch 149, with no sign of the rare-class collapse that a poorly tuned imbalance loss would cause — evidence that the $\beta = 0.999$ choice in the Class-Balanced Focal Loss was appropriate.

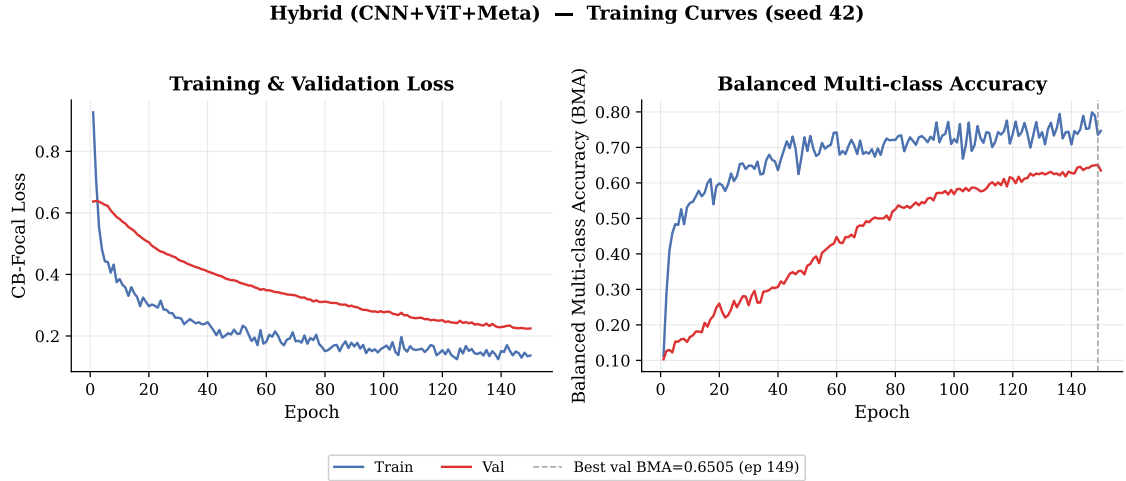


Figure 5.2: Training and validation loss (left) and BMA (right) for the lightweight hybrid model over 150 epochs. The dashed line marks the best validation BMA.

DEKAN follows a similar pattern, shown in Figure 5.3. Training and validation loss decrease smoothly over the full 150 epochs, and the validation BMA peaks late in training, again with no sign of rare-class collapse — consistent with the loss configuration transferring well from the lightweight model to the higher-capacity dual-backbone setting.

DEKAN (Dual+KAN+Meta) — Training Curves (seed 42)

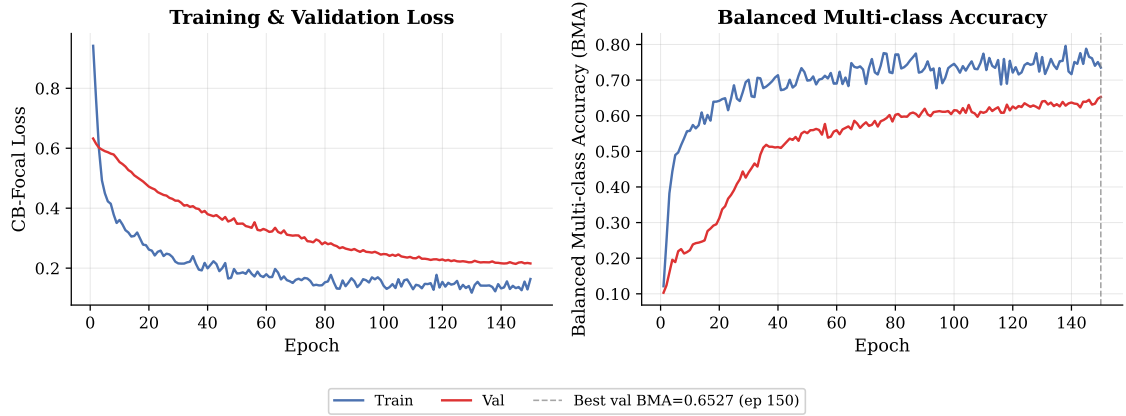


Figure 5.3: Training and validation loss (left) and BMA (right) for the DEKAN model over 150 epochs. The dashed line marks the best validation BMA.

5.2 Per-Class Analysis

Because BMA is an average over classes, it is worth looking at where the models succeed and fail per class. Table 5.2 lists per-class recall for two representative baselines and the two proposed models, and Figure 5.4 compares the lightweight hybrid against EfficientNet-B0 class by class.

Table 5.2: Per-class recall on the ISIC-2019 test set (with TTA).

Class	EfficientNet-B0	MobileViT-S	Hybrid (ours)	DEKAN (ours)
MEL	0.594	0.447	0.518	0.611
NV	0.857	0.890	0.859	0.859
BCC	0.546	0.467	0.618	0.731
AK	0.348	0.452	0.587	0.368
BKL	0.366	0.587	0.632	0.622
DF	0.245	0.660	0.566	0.509
VASC	0.843	0.882	0.863	0.922
SCC	0.157	0.430	0.223	0.529

The pattern is consistent with the headline numbers. The lightweight hybrid lifts several of the hard middle classes relative to the convolutional baselines — BCC, AK, and BKL in particular — which is where its global reasoning and metadata context help most. Notably, the hybrid’s AK recall (0.587) is the highest of any model in the study, including DEKAN (0.368); actinic keratosis is a sun-damage-related precancerous condition disproportionately common in older patients at exposed anatomical sites, precisely the profile where the metadata cross-attention shifts the prior most reliably. Its weakest class is SCC (0.223), a rare class that is easily confused with AK and BKL. DEKAN spreads its recall more evenly and recovers SCC substantially (0.529) while also posting the strongest VASC (0.922) and BCC

(0.731), which is the main reason its balanced accuracy is highest. Figure 5.5 shows the full confusion matrix for the lightweight hybrid; the off-diagonal mass sits mostly among the keratinocytic classes (AK/BKL/SCC), exactly the clinically difficult group.

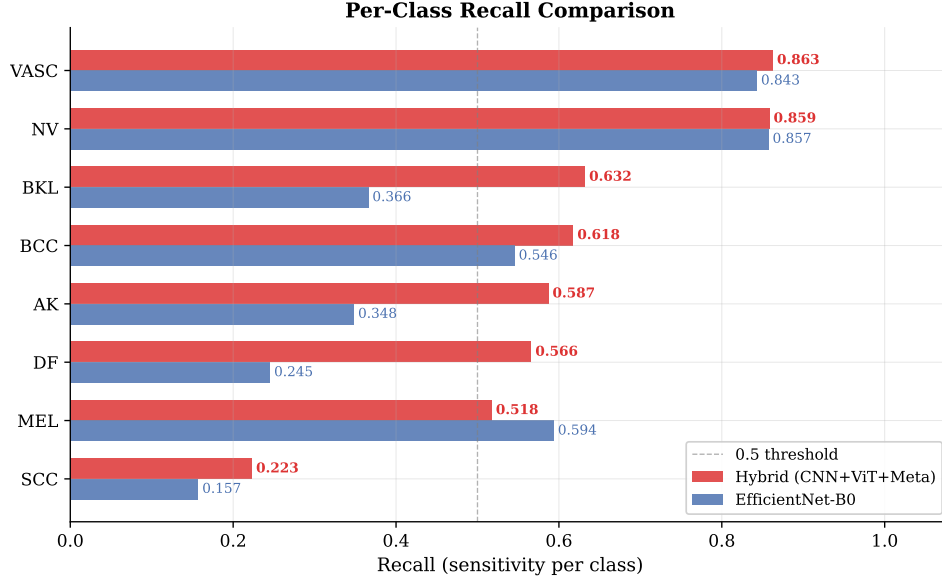


Figure 5.4: Per-class recall of the lightweight hybrid versus EfficientNet-B0, sorted by the hybrid’s recall.

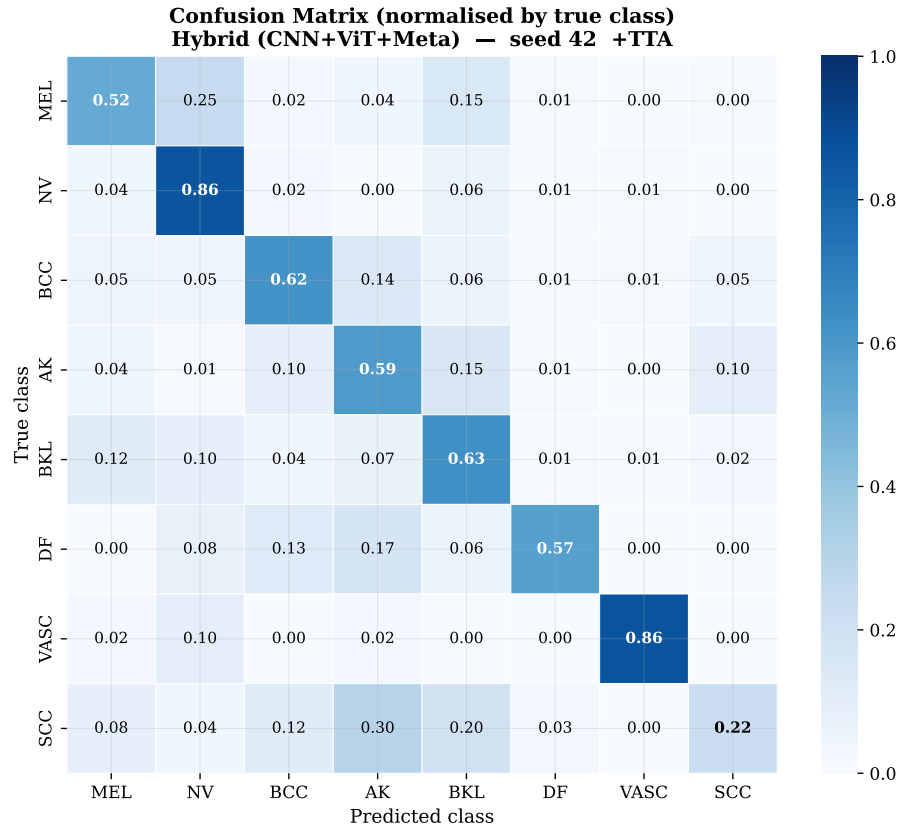


Figure 5.5: Confusion matrix for the lightweight hybrid (normalised by true class, with TTA).

Figure 5.6 shows the same matrix for the DEKAN flagship. The keratinocytic confusion (AK/BKL/SCC) is reduced relative to the lightweight hybrid, and DEKAN recovers SCC substantially (0.529 vs. 0.223) — the class that costs the hybrid most. VASC recall reaches 0.922, the highest of any model in the study.

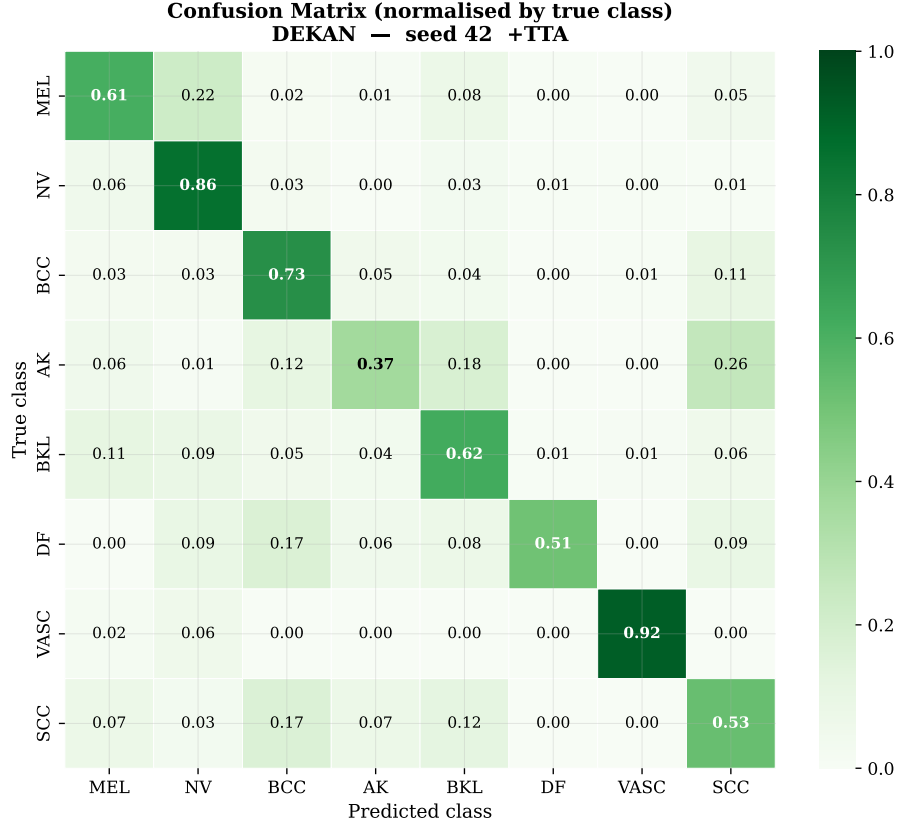


Figure 5.6: Confusion matrix for the DEKAN flagship model (normalised by true class, with TTA).

5.3 Efficiency Analysis

The point of the lightweight model is accuracy per unit of cost, so Table 5.3 reports balanced accuracy per million parameters alongside the raw numbers, and Figure 5.7 plots accuracy against parameter count with compute shown as bubble size. Among all models that fit the sub-6M budget, the lightweight hybrid has the best BMA-per-parameter (0.153). The isolated CNN-stem ablation scores higher per-parameter (0.268) and the pure-ViT ablation (0.249) do so as well, but both sit far below the full hybrid in absolute BMA, confirming that parameter-efficiency without sufficient accuracy is not a useful operating point. DEKAN sits at the opposite end: it buys the highest absolute accuracy with the lowest efficiency, which is the expected and intended trade-off between the two tiers.

Table 5.3: Efficiency: balanced accuracy per million parameters (TTA BMA).

Model	Params (M)	GMACs	BMA	BMA / M-param
ResNet-18	11.69	1.810	0.3368	0.029
MobileNetV2	3.50	0.300	0.4738	0.135
EfficientNet-B0	5.29	0.390	0.4946	0.093
MobileViT-S	5.60	1.000	0.6020	0.108
EfficientFormer-L1	12.27	1.300	0.5519	0.045
Hybrid (ours, light)	3.98	0.631	0.6081	0.153
DEKAN (ours, flagship)	16.45	6.633	0.6438	0.039

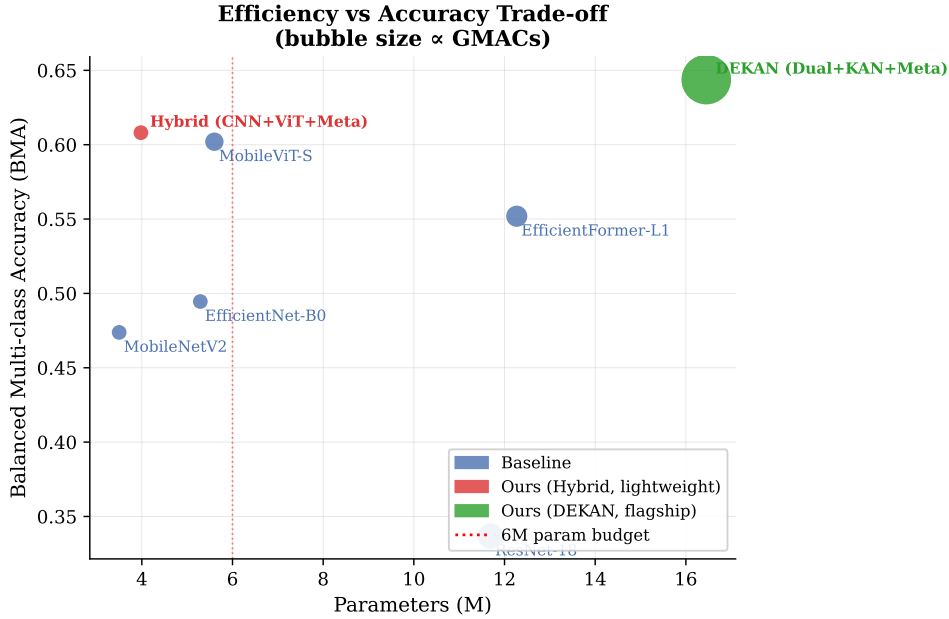


Figure 5.7: Accuracy vs. parameters; bubble size is proportional to GMACs. The dotted line marks the 6M-parameter budget. The lightweight hybrid (red) gives the best accuracy within budget.

5.4 Effect of Test-Time Augmentation

TTA helps most but not all models, so Table 5.4 reports BMA with and without it. The lightweight hybrid (full model) benefits most among the strong models (+0.027), and DEKAN gains a further +0.007. Among baselines, EfficientFormer-L1 drops slightly and ResNet-18 degrades sharply (−0.151); the ResNet-18 case is a known failure mode in which per-view predictions are confidently wrong in different directions, so averaging hurts.

The no-metadata ablation variant reveals a further pattern. Without the metadata head, the hybrid goes from 0.6216 to 0.6085 under TTA (−0.013) — the same degradation direction as ResNet-18. Adding the metadata head reverses this: the full model goes from 0.5815 to 0.6081 (+0.027). The two variants converge to nearly

the same TTA number (0.6085 vs 0.6081) from opposite sides. This TTA $\times$ metadata interaction suggests the cross-attention head stabilises per-view predictions; without it, the model becomes overconfident in different directions across augmented views.

Table 5.4: BMA without and with 8-view TTA, and the change.

Model	BMA (no TTA)	BMA (TTA)	Δ
ResNet-18	0.4875	0.3368	−0.151
MobileNetV2	0.4485	0.4738	+0.025
EfficientNet-B0	0.4917	0.4946	+0.003
MobileViT-S	0.5925	0.6020	+0.010
EfficientFormer-L1	0.5559	0.5519	−0.004
Hybrid w/o metadata	0.6216	0.6085	−0.013
Hybrid (ours, light)	0.5815	0.6081	+0.027
DEKAN (ours, flagship)	0.6368	0.6438	+0.007

5.5 Analysis of Design Solutions (Ablations)

5.5.1 Lightweight hybrid

Table 5.5 and Figure 5.8 isolate the components of the lightweight model. The pure-ViT ablation (no CNN stem, 1.97M parameters) reaches 0.4908 BMA, virtually identical to the CNN-stem-only result (0.4853). The two components are roughly equal in isolation but strongly complementary when combined: adding the ViT trunk to the CNN stem yields a gain of +0.117 to +0.123 BMA, and the full hybrid reaches 0.6081. This is the primary architectural justification for the hybrid design — neither local nor global features alone approach the performance the model achieves when both are present.

The effect of metadata in isolation is, however, modest at test time on this single run: the full model (0.6081) and the no-metadata variant (0.6085) are within 0.0004 of each other on the test set, even though on the validation set the full model led by about 0.011 BMA. We therefore report the metadata contribution for the lightweight model honestly as inconclusive on a single test evaluation; the DEKAN results below give a clearer signal. The no-TTA numbers add a structural observation: without metadata the model scores 0.6216 before augmentation (versus 0.5815 for the full model), a gap of 0.040 that is entirely closed by TTA — the full model gains +0.027 while the no-metadata variant loses −0.013 (Table 5.4). The two variants converge to the same TTA result from opposite sides, which points to the metadata head regularising per-view predictions rather than lifting the mean class probability.

Table 5.5: Lightweight hybrid ablation (TTA BMA). All four variants were trained and evaluated.

Configuration	Variant	BMA (TTA)
CNN stem + GAP + linear	hybrid_cnn_only	0.4853
Pure ViT, no CNN stem	hybrid_vit_only	0.4908
CNN + ViT, no metadata	hybrid_no_meta	0.6085
CNN + ViT + metadata (full)	hybrid_full	0.6081

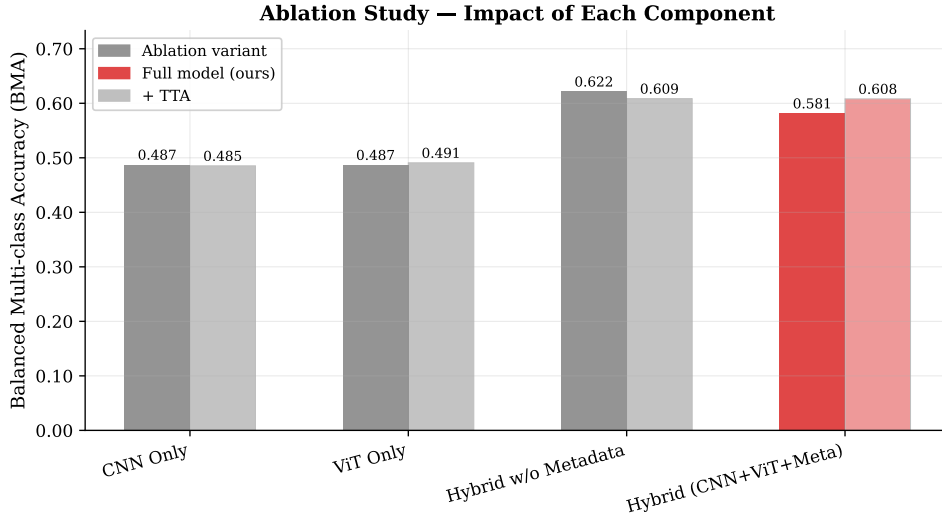


Figure 5.8: Lightweight hybrid ablation: BMA for all four variants. CNN-only and ViT-only perform similarly in isolation; the full hybrid is substantially higher.

5.5.2 DEKAN

Table 5.6 and Figure 5.9 isolate two effects in the flagship model: the contribution of clinical metadata and the choice of classifier head (KAN vs. linear).

Metadata. Comparing `dekan_no_meta` (0.6302) with `dekan_linear` (0.6510) — both use a linear classifier head, differing only in the presence of the metadata cross-attention block — shows a gain of +0.021 BMA. This is a clear and consistent signal that patient age, sex, and anatomical site provide useful diagnostic context at this model capacity.

KAN vs. linear head. Replacing the KAN classifier with a plain linear layer (`dekan_linear`, 0.6510) while keeping everything else identical produces a higher BMA than the full KAN model (`dekan_full`, 0.6438), a difference of +0.007 in favour of the linear head. The KAN classifier therefore does not improve over a plain linear layer on this task. This is an honest negative result: the KAN’s learnable splines add no measurable value at the classifier stage of a well-trained vision backbone on this noisy, imbalanced 8-class dataset. The likely explanation is that the discriminative difficulty on ISIC-2019 lies in the feature representation, not in the shape of the final

decision boundary, which a linear layer can already model adequately after the rich dual-backbone representation.

Table 5.6: DEKAN ablation: effect of clinical metadata and the KAN vs. linear classifier head (TTA BMA). The single-backbone variants were not trained, so the dual-backbone fusion gain cannot yet be quantified.

Configuration	Variant	BMA (TTA)
Dual backbone + fusion, no metadata, linear head	dekan_no_meta	0.6302
Dual backbone + fusion + metadata, linear head	dekan_linear	0.6510
Dual backbone + fusion + metadata + KAN (full)	dekan_full	0.6438

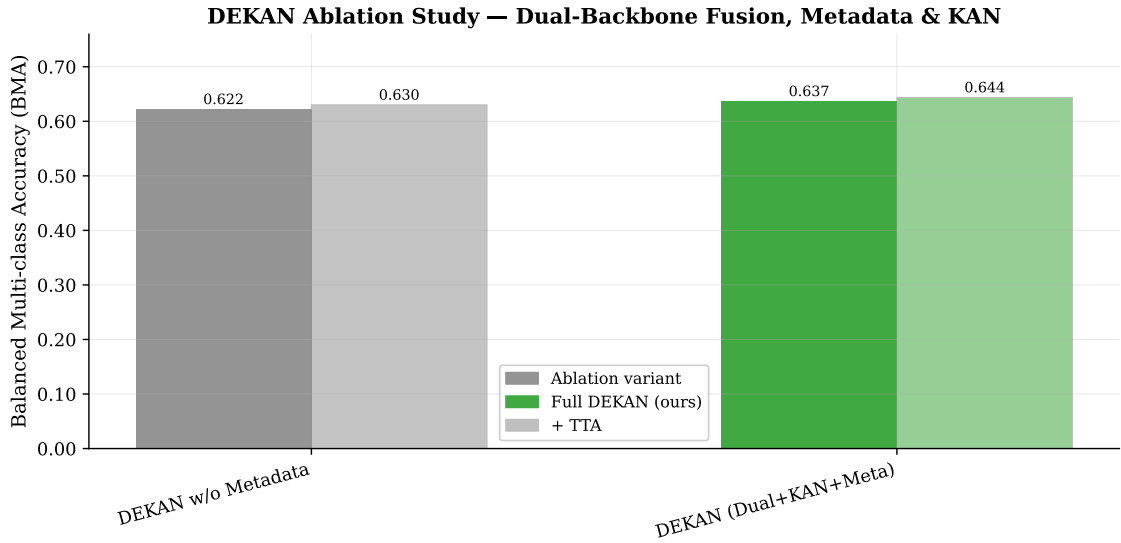


Figure 5.9: DEKAN ablation: effect of clinical metadata and KAN vs. linear classifier head on BMA. Single-backbone variants were not trained and are absent.

5.6 Comparisons and Relationships

Read against the wider literature, these results sit where one would expect a single-model, efficiency-constrained system to sit. The top entries on the ISIC-2019 challenge leaderboard [15] reach a balanced accuracy around 0.636 (DAISYLab) using large ensembles of multi-resolution EfficientNets and SENet-154. DEKAN, a single model, reaches 0.6438 BMA on our held-out split, and even the 3.98M-parameter lightweight hybrid reaches 0.6081 — within striking distance of an entire ensemble, at a tiny fraction of the cost. The comparison is not exact, since the leaderboard is computed on the official challenge test server and our split is an internal one, so we present it as context rather than as a like-for-like ranking. Relative to controlled single-model comparisons on this dataset such as Aktan et al. [26], our contribution is to add clinical-metadata fusion to the comparison and to report accuracy together with a hard efficiency budget. The clearest relationship to prior work is with the metadata-fusion line of Pacheco and Krohling [20], [22]: we adopt their core idea of representing

missing values with learned embeddings, and we extend their heavy-CNN setting to lightweight and dual-backbone hybrids.

5.7 Discussions

The results support the central claims of the thesis while leaving a few questions open. The lightweight hybrid does what it was designed to do: it beats all five same-budget baselines on balanced accuracy and is the most parameter-efficient model in its class, which makes it a credible candidate for deployment on modest hardware. DEKAN shows that the same design scales: investing capacity in a second backbone, learned fusion, and a deeper trunk pushes balanced accuracy to the level of a leaderboard ensemble in a single model. Metadata fusion helps clearly in the higher-capacity model and is, at worst, harmless in the lightweight one.

Two findings cut against initial expectations and are worth stating plainly. First, the benefit of metadata for the lightweight model could not be confirmed on a single test evaluation, even though the validation curve favoured it by 0.011 BMA. The DEKAN results give a clearer signal at higher capacity (+0.021 BMA between `dekan_no_meta` and `dekan_linear`), suggesting the metadata effect requires sufficient model capacity to manifest consistently. Second, the KAN classifier in DEKAN does not outperform a plain linear head: `dekan_linear` (0.6510) beats `dekan_full` (0.6438) by 0.007. This is the highest single-model BMA achieved in the study, but it belongs to the linear-head ablation, not to the KAN model, which is the intended flagship. The KAN’s expressive spline activations appear to offer no advantage over a linear classifier once the dual-backbone representation is rich enough.

5.8 Limitations

Several limitations bound these conclusions. All numbers come from a single seed (42); publication-grade claims would require repeating each run over multiple seeds and reporting mean and standard deviation, which we have not yet done. Two ablation variants were not trained — `dekan_densenet_only` and `dekan_effnet_only` — so the individual contribution of each backbone and the precise size of the dual-backbone fusion gain remain unquantified. We also did not evaluate on an external dataset, so the generalisation of these models beyond the ISIC-2019 distribution is untested. Finally, the comparison with the ISIC-2019 leaderboard is contextual, not a controlled head-to-head, because the evaluation splits differ.

Chapter 6

Conclusion

6.1 Summary

This thesis set out to build skin-lesion classifiers for ISIC-2019 that are both accurate across all eight classes and practical to deploy, and that use the clinical metadata a dermatologist would normally have on hand. We approached the problem with a single design philosophy — convolutional features for local texture, a transformer trunk for global structure, and a cross-attention head that folds in age, sex, and lesion site — realised at two points on the efficiency curve. The lightweight hybrid, at 3.98M parameters and 0.631 GMACs, reaches a balanced multi-class accuracy of 0.6081 and is the most parameter-efficient model in the sub-6M budget, beating all five lightweight baselines including the strong hybrid baselines MobileViT-S and EfficientFormer-L1. The flagship DEKAN model, at 16.45M parameters, reaches 0.6438 BMA, comparable to the balanced accuracy of the top ISIC-2019 leaderboard ensemble in a single model; a linear-head ablation of the same architecture (`dekan_linear`) reaches 0.6510, the highest figure in the study, with the implication that the KAN classifier head adds no accuracy benefit on this task.

6.2 Contributions

The main contributions of this work are: (1) a lightweight hybrid CNN–Transformer with clinical-metadata cross-attention that handles missing fields with learned embeddings and meets an explicit budget of under 6M parameters and 1 GMAC; (2) DEKAN, a higher-capacity model in the same family that fuses two complementary pretrained backbones through learned attention and adds a Kolmogorov–Arnold classifier head; (3) a fair, single-protocol comparison of both models against five established baselines, with controlled ablations that quantify what each component adds, including the finding that neither a CNN stem nor a ViT trunk alone approaches the performance of their combination; and (4) two honest negative or inconclusive results — that the metadata benefit for the lightweight model could not be confirmed on a single test-set evaluation (though DEKAN gives a clear +0.021 BMA signal at higher capacity), and that the KAN classifier head in DEKAN does not improve over a plain linear head (`dekan_linear` at 0.6510 beats `dekan_full` at 0.6438 by +0.007).

6.3 Future Work

The most immediate next steps follow directly from the limitations in Chapter 5. Repeating every run across at least three seeds and reporting mean and standard deviation would put the comparisons on a publication-grade footing. Completing the two remaining unfinished ablations — DenseNet-121 alone and EfficientNet-B0 alone within the DEKAN framework — would pin down the individual contribution of each backbone and quantify the size of the dual-backbone fusion gain. Evaluating on an external dataset, for example mapping the eight-class output to a melanoma-versus-rest decision on a separate collection, would test generalisation beyond ISIC-2019. Finally, qualitative interpretability — Grad-CAM on the CNN stem and attention rollout on the transformer trunk — would help confirm that the models attend to clinically meaningful regions, which matters for any tool intended to support, rather than replace, a clinician.

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Appendix A: Reproducibility

Dataset. ISIC-2019 training collection (25,331 images, 8 classes), reduced to 24,068 images after perceptual-hash deduplication and split 70/10/20 (16,872 / 2,400 / 4,796) with a lesion-grouped, class-and-source stratified split.

Preprocessing. Shades-of-Gray colour constancy ($p = 6$), short-side resize and centre-crop to 224×224 , stored as a single memory-mapped `uint8` array.

Loss and optimisation. Class-Balanced Focal Loss ($\beta = 0.999$, $\gamma = 2.5$, label smoothing 0.1); AdamW (base LR 3×10^{-4} , $\beta = (0.9, 0.999)$, gradient clip norm 5); 5-epoch linear warmup then cosine annealing to 10^{-6} ; weight EMA (decay 0.9998); weighted random sampler; automatic mixed precision and channels-last layout. Augmentation: flips, random resized crop, colour jitter, RandAugment, and Mixup ($\alpha = 0.4$).

Per-model batch / accumulation / epochs. See Table 4.1. The lightweight hybrid uses batch 96, accumulation 3, 150 epochs; DEKAN uses batch 64, accumulation 4, 150 epochs.

Evaluation. Primary metric BMA (mean per-class recall); secondary macro-F1, macro-AUC, accuracy; 8-view TTA (flips and rotations, mean softmax); checkpoint selected by best validation BMA. All reported numbers use seed 42.

Figures. Figures in this thesis are generated from the saved evaluation outputs. The expected figure files are listed below; the architecture schematic (Figure 4.1) is drawn inline with TikZ and needs no external file.

- 01_training_curves.pdf; 02_model_comparison.pdf
- 03_confusion_matrix.pdf; 04_per_class_recall.pdf
- 05_efficiency_scatter.pdf; 06_ablation.pdf
- 07_dekan_ablation.pdf; 08_confusion_matrix_dekan.pdf
- 09_training_curves_dekan.pdf
- A1_baseline_confusion_matrices.pdf (Appendix B)
- A2_all_training_curves.pdf (Appendix B)

Appendix B: Extended Results

Full Per-Class Recall (Test Set, with TTA)

Table 6.1 gives per-class recall for all five baselines plus four key variants (the pure-ViT ablation, the proposed lightweight hybrid, the linear-head DEKAN ablation, and the full DEKAN), allowing direct column-by-column comparison. The BMA row repeats the headline figure from Table 1 for reference. Proposed models and their ablation variants are marked †.

Table 6.1: Per-class recall on the ISIC-2019 test set (8-view TTA, seed 42). † = proposed model or direct ablation variant.

Class	ResNet-18	MobileNetV2	Eff-B0	MViT-S	EFormer-L1	ViT Only†	Hybrid†	DEKAN (Linear)†	DEKAN†
MEL	0.231	0.511	0.594	0.447	0.572	0.464	0.518	0.600	0.611
NV	0.841	0.837	0.857	0.890	0.892	0.785	0.859	0.841	0.859
BCC	0.085	0.653	0.546	0.467	0.734	0.450	0.618	0.752	0.731
AK	0.523	0.323	0.348	0.452	0.252	0.439	0.587	0.439	0.368
BKL	0.057	0.446	0.366	0.587	0.523	0.250	0.632	0.583	0.622
DF	0.189	0.132	0.245	0.660	0.321	0.340	0.566	0.679	0.509
VASC	0.216	0.706	0.843	0.882	0.882	0.843	0.863	0.843	0.922
SCC	0.554	0.182	0.157	0.430	0.240	0.355	0.223	0.471	0.529
BMA	0.337	0.474	0.495	0.602	0.552	0.491	0.608	0.651	0.644

Training Curves — All Models

Figure 6.1 plots validation and training BMA over epochs for all seven main models. The dotted vertical line marks the best-validation checkpoint for each model.

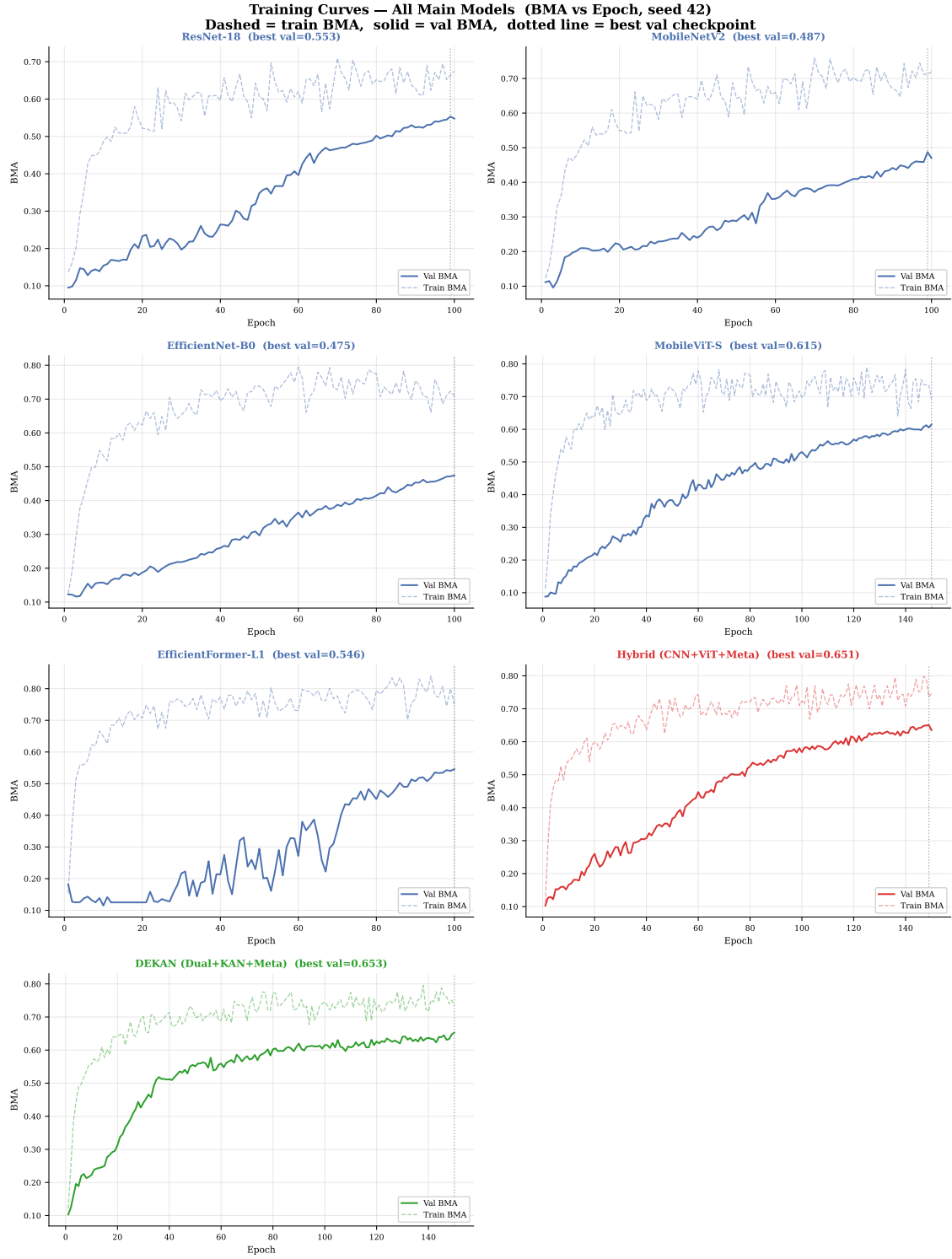


Figure 6.1: Validation BMA (solid) and training BMA (dashed) for all five baselines and both proposed models (seed 42). Proposed models are shown in colour (red = lightweight hybrid, green = DEKAN); baselines in blue.

Confusion Matrices — Baseline Models

Figure 6.2 shows normalised confusion matrices for all five baseline models. All matrices use TTA predictions except ResNet-18, for which only the non-TTA confusion matrix was saved (consistent with the per-class recall numbers in Table 6.1).



Figure 6.2: Confusion matrices (normalised by true class) for the five baseline models. The keratinocytic classes (AK, BKL, SCC) are the primary source of off-diagonal mass across all baselines.